

	Application scope	Breathing type	Training objective	Interaction design	Feedback modality	Haptic actuator	Actuator placement	Haptic pattern, frequency and intensity	Sensor measurement	Sensor placement	Timing of feedback	Frequency of feedback	Function of feedback	KP or KR	Qualitative or quantitative	Single error or multi error response	Prescriptive (what should have happened) or descriptive	Reinforcement type	Focus of attention
Parnandi	Therapeutic, well-being and stress-management	BR	Learning, engagement	adaptive	Visual	-	-	-	Breathing rate with expansion and contraction	Chest	Concurrent	Bandwidth				Single			
Abushakra		breath depth, inhale/exhale phases, and lung capacity estimation	Engagement	Adaptive	Visual, audio	-	-	-	Breathing sound	Mouth	Concurrent, terminal	Average	Motivating and reinforcing		Qualitative	Single and multi	Descriptive		
Choi	Therapeutic, well-being and stress-management	BR, breathing direction	Engagement, performance	Functional	Haptic	3 soft-air pouches	Torso	Tactile pattern (synchronized or sequential), Intensity level (adjustable air pressure) , Feedback frequency (matching a goal breathing rate)	-	-	Concurrent	* Self-selection	Guiding	-	-	-	Prescriptive		
Dijk	Therapeutic, well-being and stress-management	BR, breathing direction	Engagement	Adaptive, functional	Haptic, audio, visual	176 Vibration motors	Full-body	1) predefined pattern, intensity, and frequency. 2) only adjusts pattern to user's breathing. 3) Adjusts Pattern, Intensity, and Breathing Rate (HRV-based optimization)	1) none. 2) respiration rate. 3) respiration rate and HRV	2) abdomen. 3) abdomen and finger	Concurrent, terminal	-	Motivating, guiding, reinforcing	KP	Qualitative	Single, multi	Both	Positive reinforcement	Internal, external
Greinacher	Sports and performance enhancement	BR, Breathing rhythm	Performance, learning	Adaptive	Haptic, audio, visual	8 Vibration motors	Lower back	pattern is adaptive. Frequency is predefined. Intensity is predefined.	breathing pattern	chest	Concurrent	-	Guiding, motivating, reinforcing, informing	KP	Qualitative	Single	Prescriptive	Positive reinforcement	Internal
Jung	Therapeutic, well-being and stress-management	Diaphragmatic breathing, breathing rhythm	Engagement	Adaptive, functional	Haptic	2 Inflatable pads	Upper and lower back	pattern can be applied symmetrically or asymmetrically, synchronous or asynchronous. Frequency predefined based on interaction mode. Intensity dynamically adjusted based on user's breathing	displacement of chest or stomach	chest or stomach	Concurrent	Bandwidth, fading, self-selection	Guiding, Attuning, Informing, Reinforcing	KP	Qualitative	Single	Both	Positive, mildly negative	Internal
Karpashevich	Therapeutic, well-being and stress-management	BR, breathing rhythm	Engagement, learning	Adaptive	Haptic, audio	10 Inflatable pillows	chest, lower abdominal muscles, left and right of ribcage, lower back	pattern predefined and adaptive (depend on exercise): in-phase and anti-phase feedback. Frequency predefined. Intensity manually adaptable	expansion and contraction	torso	Concurrent, terminal	Bandwidth, fading, average	Guiding, informing, attuning, motivating, reinforcing	KP	Qualitative	Single, multi	Both	Positive	Internal
Liang	Therapeutic, well-being and stress-management	BR, breathing depth	Engagement, performance	Adaptive	Haptic, visual	Tactile nad inertia through fidget spinner	Hand	Pattern adaptive responding to breath control. Frequency based on fidget spinner's rotation. Intensity based on force of exhalation of user	HRV, respiration rate	Finger	Concurrent, terminal	Bandwidth, average	Guiding, informing, attuning, motivating, reinforcing	KP	Both	Single	Prescriptive	Positive	External
Loriga		Diaphragmatic breathing, thoracic breathing	-	Non-interactive	-	-	-	-	Heart electrical activity, respiratory activity, breathing patterns, voltage variations due to thoracic impedance changes	Wearable garment, thorax, abdomen	-	-	-	-	-	-	-	-	-
Prpa2016	Therapeutic, well-being and	Diaphragm breathing	Learning, engagement	Adaptive	Visual, audio	-	-	-	Brain activity, respiration rate	Thorax, diaphragm	Concurrent, terminal	Bandwidth, average, fading	Reinforcing, motivating	KP	Qualitative	Multi, single	Descriptive	Positive	Internal
Qin	Medical and clinical	BR, breathing rhythm, deep breathing	Learning, engagement	Adaptive	Haptic, visual, auditory	unclear	unclear	unclear	Respiration rate, waveform and amplitude	Chest, abdomen	Concurrent, terminal	Bandwidth, average, summary	Motivating, reinforcing, guiding, informing	kp	qualitative, some quantitative	multi, single	Prescriptive	positive	internal
van Rooij	Therapeutic, well-being and	BR, diaphragmatic breathing	Learning, engagement	Adaptive	Visual	-	-	-	expansion and contraction	diaphragm	Concurrent	Bandwidth	Motivating, attuning, reinforcing	KP	qualitative	Multi, single	Descriptive	Positive	External
Sefidgar		BR	Engagement	Functional	Haptic	Moving cage, vibrato in motor, stiffening actuator	In hand	Pattern, frequency and intensity are predefined	Touch sensor	Entire 'animal'	Concurrent	-	Guiding, informing, attuning	-	-	-	-	-	External
Shameki	Therapeutic, well-being and stress-management	BR, deep breathing	Engagement	Adaptive	Visual, auditory	-	-	Deep breath-based feedback, respiration mirroring	Respiration rate, chest expansion	Chest	Concurrent	Bandwidth, average	Guiding, motivating, attuning	KP	Qualitative	Single	Descriptive	Positive	Internal, external
Theodore	Therapeutic, well-being and stress-management	BR, breathing rhythm	Engagement	Adaptive	Haptic	Linear actuators	Center of cushion	Pattern integrates user's breath mirroring with predetermined progressive guidance	Barometric pressure sensor (BMP280)	Abdomen (inside airtight airbag)	Concurrent	Bandwidth, fading	Guiding, attuning, reinforcing	KP	Qualitative	Multi	Descriptive	Positive	Internal
Tsakanaki	Therapeutic, well-being and stress-management	BR, diaphragmatic breathing, muscle activation	Engagement, learning	Adaptive	Haptic	Shape-changing pillows	Torso	Shape-change sensor-actuator material provides tactile feedback by inflating against the body to create awareness	EMG sensor for rectus abdominis contraction, strain sensor for breathing movement	Torso, rectus abdominis muscle, intercostals	Concurrent	Bandwidth, fading	Guiding, attuning, reinforcing	KP	Qualitative	Multi	Descriptive	Positive	Internal
Wang	Therapeutic, well-being and stress-management	BR, breathing rhythm	Engagement, learning	Adaptive	Haptic	10 vibration motors	Torso (front and back)	Predefined patterns with adjustable intensity and frequency variation for relaxation, calmness, and focus	HRV, respiration rate	Chest, abdomen	Concurrent	Bandwidth, average	Guiding, attuning, motivating, reinforcing	KP	Qualitative	Multi	Prescriptive	Positive	Internal
Yang	Therapeutic, well-being and stress-management	Thoracic and abdominal breathing	Learning, engagement	Adaptive	Haptic	Capacitive textile sensors	Upper chest and lower abdomen	Haptic pattern based on breathing phase variations, frequency varies with breathing rate	Thoracic and abdominal breathing signals	Chest, abdomen	Concurrent	Bandwidth, adaptive	Guiding, informing, reinforcing	KP	Qualitative	Multi	Prescriptive	Positive	Internal
Yu	Therapeutic, well-being and stress-management	BR	Engagement, learning	Adaptive	Haptic	Shape-changing airbag	Hand	Expansion and contraction simulating breathing movements, synchronized with breathing rhythm	PPG, respiration rate	Finger, abdomen	Concurrent	Bandwidth	Guiding, attuning, reinforcing	KP	Qualitative	Multi	Prescriptive	Positive	Internal